

Intro to Econ 628

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UBC

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Why is this course important?

Technology and Big Data Are Changing Economics: Mining Text to Track Methods

Janet Currie, Henrik Kleven, and Esmée Zwiers

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ABSTRACT

The last 40 years have seen huge innovations in computing technology and data availability. Data derived from millions of administrative records or by using (as we do) new methods of data generation such as text mining are now common. New data often requires new methods, which in turn can inspire new data collection. If history is any guide, some methods will stick and others will prove to be a flash in the pan. However, the larger trends towards demanding greater credibility and transparency from researchers in applied economics and a “collage” approach to assembling evidence will likely continue.

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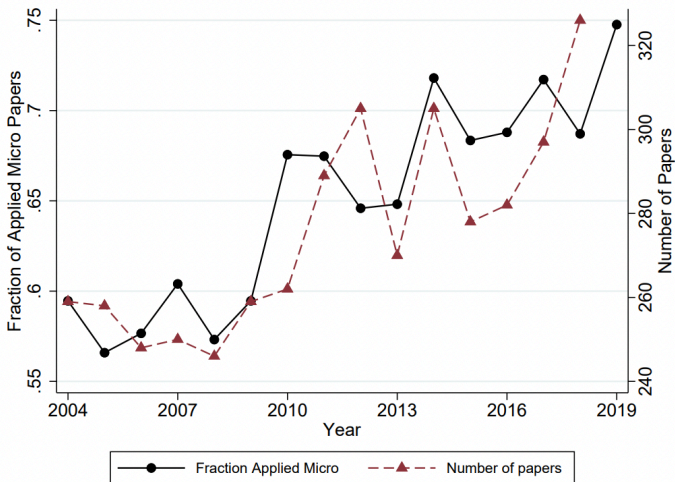
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Why is this course important?

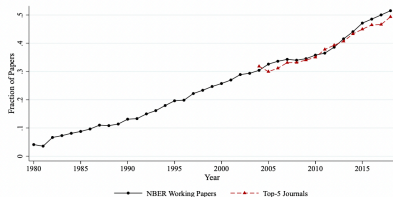
Figure I: Fraction of Applied Microeconomics Articles in Top-5 Journals



Why is this course important?

Figure II: The Credibility Revolution

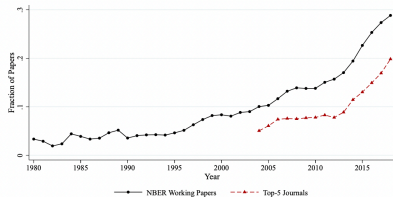
A: Identification



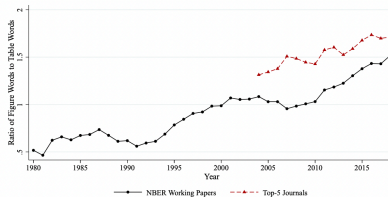
B: All Experimental and Quasi-Experimental Methods



C: Administrative Data



D: Graphical Revolution

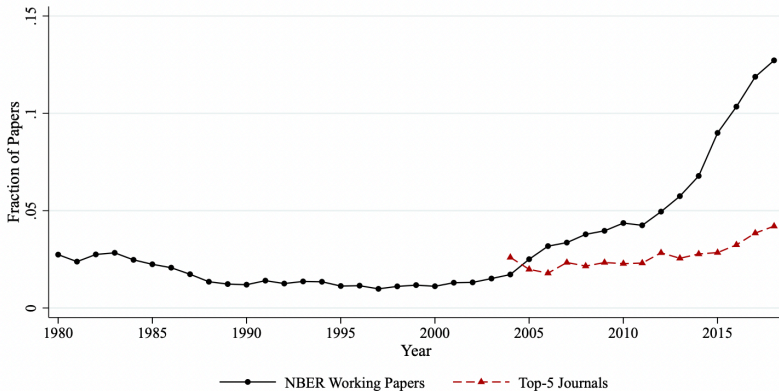


Notes: This figure shows different dimensions of the “credibility revolution” in economics: Identification (Panel A), All Experimental and Quasi-Experimental Methods (Panel B), Administrative Data (Panel C), and the Graphical Revolution (Panel D). Panel D shows the ratio of the number of ‘Figure’ terms mentioned to the number of ‘Table’ terms mentioned. See Table A.1 for a list of terms. The series show 5-year moving averages.

Why is this course important?

Figure III: Experimental Methods

A: RCTs

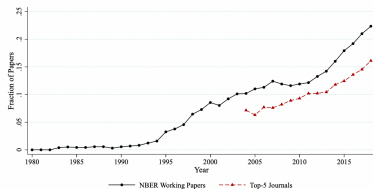


B: Lab Experiments

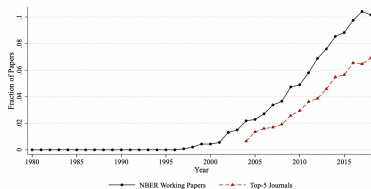
Why is this course important?

Figure IV: Quasi-Experimental Methods

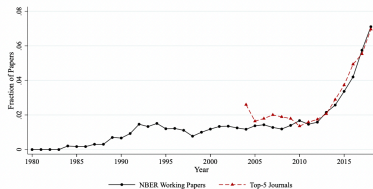
A: Difference-in-Differences



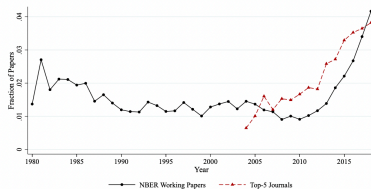
B: Regression Discontinuity



C: Event Study



D: Bunching



Notes: This figure shows the fraction of papers referring to each type of quasi-experimental approach. See Table A.1 for a list of terms. The series show 5-year moving averages.

More recent numbers

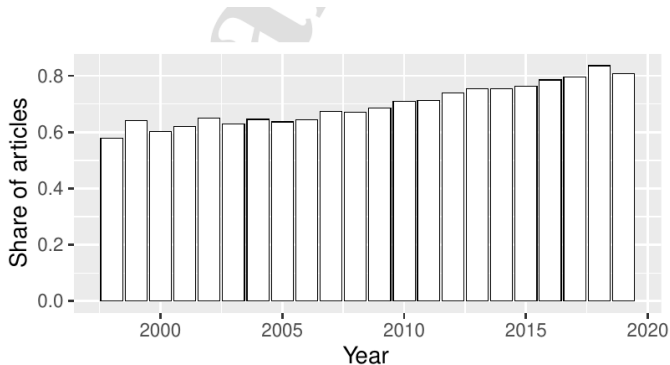
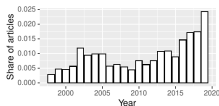
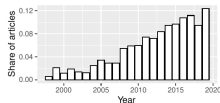


Figure 3: Share of empirical articles per year (all articles: 31k)

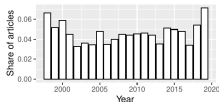
More recent numbers



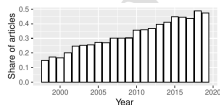
(a) Bunching



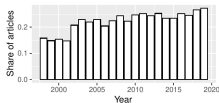
(b) Difference-in-differences



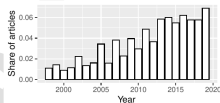
(c) Event study



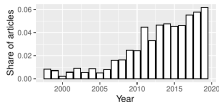
(d) Identification



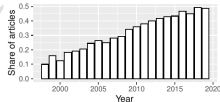
(e) Instrumental variables



(f) Lab experiment



(g) Field experiment



(h) Fixed effects

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- But empirical work is hard
 - Fundamentally messy.
 - No clear start (statement of a theorem) and end (proof of the theorem).
 - Data is always messy $\rightarrow$ discretion will always come into play.
 - There is no Q.E.D at the end of empirical papers.

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This can be overwhelming for grad students entering their third year.

- They need to do research *independently*.
- Often this research is empirical.
- But most of your prior training was just about proving theorems...

Econ 628 is designed to smooth this transition

Section 1

A primer to conduct empirical research

A few tips

- Some guidelines on how to keep the “mess” manageable when working with data.
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- Some guidelines on how to keep the “mess” manageable when working with data.
- Many of these tips were passed on to me by Patrick Kline who taught the equivalent of Econ 628 at Berkeley.
- Every empirical project can be divided into three phases
 - 1 Idea.
 - 2 Build.
 - 3 Analysis.
 - 4 Marketing.

Skill Sets

- Each stage requires its own set of skills
- Idea: creativity, familiarity with existing literature, knowledge of institutional details, econometric skills (“see a research design”)
- Build: attention to detail, coding skills, lots of patience.
- Analysis: understanding of economics, econometrics and the linkage of the two, efficiency in summarizing key patterns and ruling out others all with a commitment to scientific objectivity.
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- Marketing: writing and presentation of your work.
- Increasing common to work in “teams”.

The Idea stage

- Best empirical projects: “shift priors”.
- Good empirical papers provide facts while ruling out other patterns in a convincing way.
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- Some general tips:

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- Good empirical papers provide facts while ruling out other patterns in a convincing way.
- Also needs to be consequential (“Macro” angle).
- Some general tips:
 - *Know your audience.*
 - *One side test projects are risky.*
 - *Treatment must be measured accurately.*
 - *Bias/Variance trade offs matter.*
 - *Focus on your comparative advantage.*

The Build stage

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- Every empirical project must have a script called “build”
- A build script:
 - Reads raw data, cleans it, save a “analysis” file.
 - Nothing glamorous about this but immensely important.
 - You can spend 2 years of grad school just working on a build script.
THIS IS NORMAL
- Some tips:
 - *Check your raw data*
 - *NEVER WORK IN THE COMMAND WINDOW*
 - *Keep log files*
 - *Test and verify your knowledge of the data*
 - *Store your data somewhere safe (learn Github and/or use Dropbox)*
 - *Chill.*

The Analysis stage

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- Some tips
 - *Start simple.*
 - *Change one thing at the time (and keep track of them).*
 - *Automate tables.*
 - *Aesthetic matters.*
 - *Check magnitudes.*
 - *Always report number of observations.*
 - *Tell the truth.*

Marketing Stage

- This is the final phase of your empirical project. It is when you are going to communicate your results via either a paper or some presentation.
- Vast majority of students underestimate the importance of this stage.
- Being able to communicate in simple and plain words the key core idea of your paper is crucial.
- Most JM talks / paper rejection occur because presentation / writing is not clear.

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- Most JM talks / paper rejection occur because presentation / writing is not clear.
- *Make nice to read tables and figures. Reader be able to understand what is going without having to read the entire paper..*
- *Attend all seminar presentations in your field.*
- *Practice a lot your JM talk.*
- *Pass the grandma test.*

Pop-Quiz! From Dynarski et al. (2021, AER)

1740

THE AMERICAN ECONOMIC REVIEW

JUNE 2021

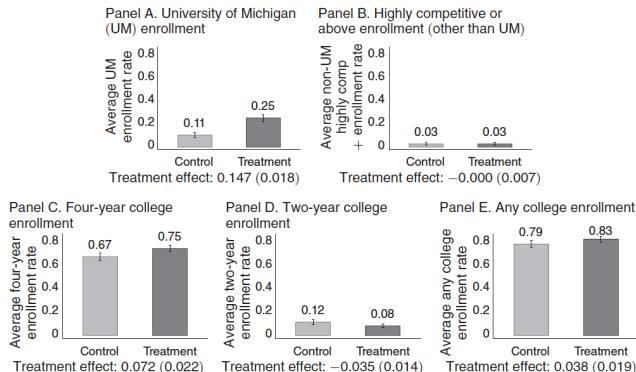


FIGURE 6. ESTIMATED EFFECT OF HAIL SCHOLARSHIP ON COLLEGE CHOICE, FIRST AND SECOND HAIL COHORTS

Notes: All analyses done at the school-year-level. Ninety-five percent confidence intervals shown based on standard errors clustered at the school level. Enrollment measured at the first institution attended in the fall following expected high school graduation. Enrollment unconditional on any college enrollment. Treatment effects estimated from a regression of the outcome on an indicator for treatment status and strata dummies. Robust standard errors clustered at the school level reported in parentheses.

Source: Michigan administrative data

One more... (this is from MHE)

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Table 3.2.1: Estimates of the returns to education for men in the NLSY

	(1)	(2)	(3)	(4)	(5)
Controls:	None	Age dummies	Col. (2) and additional controls*	Col. (3) and AFQT score	Col. (4), with occupation dummies
	0.132 (0.007)	0.131 (0.007)	0.114 (0.007)	0.087 (0.009)	0.066 (0.010)

Notes: Data are from the National Longitudinal Survey of Youth (1979 cohort, 2002 survey). The table reports the coefficient on years of schooling in a regression of log wages on years of schooling and the indicated controls. Standard errors are shown in parentheses. The sample is restricted to men and weighted by NLSY sampling weights. The sample size is 2434.

*Additional controls are mother's and father's years of schooling and dummy variables for race and Census region.

Last one... (this is from Borjas, 1980)

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TABLE 1
SELECTED WAGE COEFFICIENTS
DEPENDENT = $\ln(H_w)$

Measure of Wage (in logs)	Sample			
	Pooled	Hourly	Weekly	Annual
Usual wage rate	-.0383 (-2.37)	.0664 (1.51)	-.1040 (-3.81)	-.0256 (-1.11)
Predicted wage rate	.0269 (.59)	.1562 (1.47)	.0596 (.22)	.0863 (1.30)
Sample size	1561	278	449	834

Note: *t*-ratios are given in parentheses in all tables of this paper. Held constant in the regression are: other nonwage income in the family, expected number of years until retirement, years of labor force experience, number of children, whether job information refers to current or last (if not currently working) job, marital status, health status, and education.

Section 2

What is Econometrics?

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① Identification

② Estimation

③ Inference

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What is econometrics?

- 1 Identification (What can be learned?)
 - 2 Estimation (How do I learn it?)
 - 3 Inference (How certain should I be?)
- Econometricians: new results in these areas.
 - Applied Economists: formulate hypothesis, apply tools 1-3 created by econometricians to real data.

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- Good question. Some differences:
 - Applications to economic questions / data.
 - Emphasis on causal relationship using observational data.
 - Impose restrictions from economic theory.
 - Emphasis on welfare and counterfactual distributions.

Section 3

Econometric Models and Causality

Roadmap

- Today: Establish a link between basic concepts in econometric models and their relationship with modern notions of causality.
- For a long time in economics causality could be achieved/discussed only *after* having specified an econometric model.
- An econometric model must have two fundamental features:
 - It tells us how the observed data were generated.
 - It tells us how things would have been if we were to exogenously change one factor *ceteris paribus*
- Econometric models impose structure on the data generating process (DGP)
- Knowledge of the DGP yields a probability measure $\mathbf{P}(\cdot)$ over the observed data $\mathbf{Y}$.

- 1 Parametric models: models are indexed by a finite dimensional vector of parameters.
- 2 Non-parametric models: models are indexed by infinite dimensional parameters (e.g. an unknown function).
- 3 Semi-parametric models: models are indexed by a finite dimensional vector parameters and an infinite dimensional nuisance function.

A parametric model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad (1)$$

$$(X_i, \varepsilon_i) \sim_{\text{iid}} \mathcal{N} \left[\begin{pmatrix} \mu \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma^2 \end{pmatrix} \right] \quad (2)$$

$$\theta = (\beta_0, \beta_1, \mu, \log \sigma_x, \log \sigma) \in \mathcal{R}^5 \quad (3)$$

- Everything there is to know about the world in 5 parameters.

A semi-parametric model

$$Y_i = \beta X_i + \varepsilon_i \quad (4)$$

$$\mathbb{E}[\varepsilon_i | X_i] = 0 \quad (5)$$

$$\beta \in \mathcal{R} \quad (6)$$

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- Unrestricted distribution of $F_X(\cdot)$ of X_i .
- Mean restriction on conditional distribution of $F_{\varepsilon|X}$
- Both objects potentially infinite dimensional (but nuisance?)

A non-parametric model

$$Y_i = g(X_i, \varepsilon_i) \quad (7)$$

$$\varepsilon_i \perp X_i \quad (8)$$

$$g(., .) : \mathcal{R}^2 \rightarrow [0; 1] \text{ is monotone and increasing in both arguments} \quad (9)$$

- Unrestricted marginals of ε and X_i .
- Interested in function $g(., .)$ or features such as $E_\varepsilon[g(X_i, \varepsilon_i)]$.

Identification

- Identification is the study of which DGPs in the model space can be ruled out given the joint distribution of observables.
- This question is asked in world where DGPs can be estimated infinitely many times (that is never confuse identification with estimation).
- Logical first step for any empirical project.
- If nothing is identified, not worth going into step 2 and 3 (estimation, inference).

Setup

- Most models can be written as

$$\mathbf{Y}_i = \mathbf{g}(\mathbf{U}_i), \quad \mathbf{U}_i \sim_{iid} F_U(\mathbf{u}) \quad (10)$$

- We call $\theta \equiv (\mathbf{g}(\mathbf{U}_i), F_U)$ the structure of the model.
- $F_Y(\mathbf{y})$ the distribution function governing the observed variables $\rightarrow$ e.g. what is observed mean Y in my data.
- $F_\theta(\mathbf{y})$ the distribution function implied by θ under the model $\rightarrow$ e.g. what is the mean Y for a particular configuration of θ .

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- Two structures θ' and θ'' are observationally equivalent if $F_{\theta'}(.) = F_{\theta''}(.)$

- The identified set is given by

$$\Omega(F_Y, \Theta) = \{\theta \in \Theta : F_\theta(.) = F_Y(.)\} \quad (11)$$

- Structure is point identified is $\Omega(F_Y, \Theta)$ is a singleton.

Example: A Mixture Model

Consider the model

$$Y_i = (1 + D_i) \varepsilon_i, \quad (D_i, \varepsilon_i) \stackrel{iid}{\sim} \text{Bernoulli}(p) \times N(0, 1), \quad p \in (0, 1)$$

where we observe data on Y_i .

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Proof.

By law of total probability:

$$f_Y(y) = \phi(y)(1-p) + \frac{1}{2}\phi\left(\frac{y}{2}\right)p$$

Solving for p , we have $p = \frac{f_Y(y) - \phi(y)}{\frac{1}{2}\phi\left(\frac{y}{2}\right) - \phi(y)}$



- How many moments in the data can help you identify p ?

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- This question is the foundation of over-identification tests. We will go back to this concept many times!

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$$\Delta_i(x'', x') \equiv f(x'', U_i; \phi) - f(x', U_i; \phi)$$

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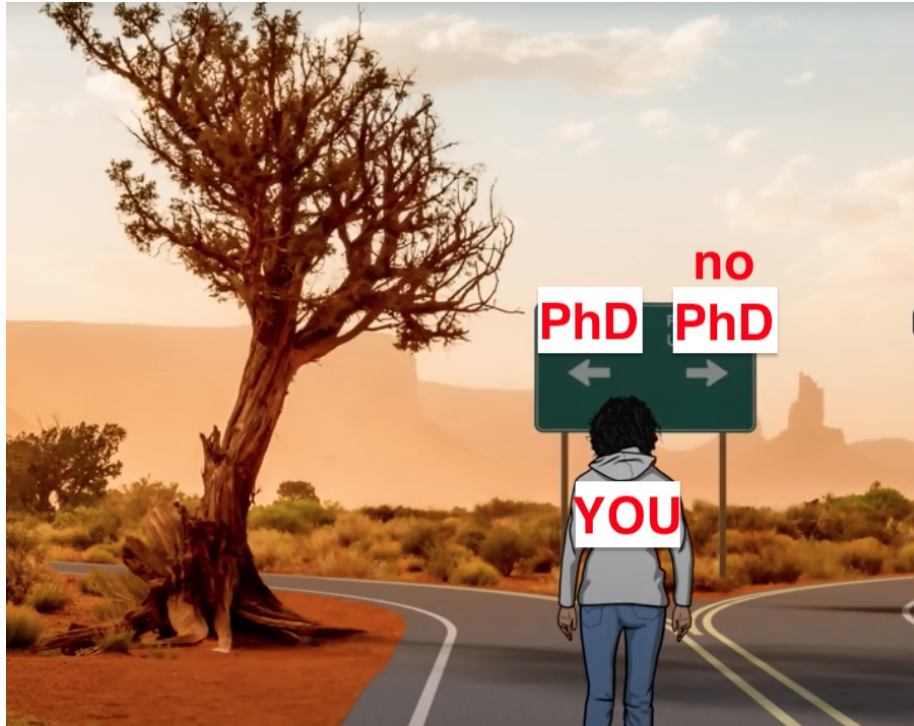
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- Causal effect of changing X_i from x' to x'' :

$$\Delta_i(x'', x') \equiv f(x'', U_i; \phi) - f(x', U_i; \phi)$$

- Need to be careful: can we change X_i holding $f(., .)$ and U_i constant?
- More generally, what if my model is wrong?



Potential Outcomes

- Statistics approach (Neyman-Rubin): Define counterfactuals “atheoretically” relative to what would emerge from an experiment.
- Potential outcome $Y_i^d \equiv Y_i(d)$ gives what results when unit i is experimentally exposed to treatment d

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 - D_i : 1 if treated, 0 if placebo
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 - D_i : 1 if treated, 0 if placebo
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- “Fundamental problem of causal inference” (Holland, 1986)

$$Y_i = D_i Y_i^1 + (1 - D_i) Y_i^0$$

Unconfoundedness and ATE

i	Y_i^0	Y_i^1	D_i
1	32	20	1
2	200	4	0
3	0	0	0
4	80	80	1
5	53	88	0
6	0	90	1

- Causal inference requires restrictions on the “missing” potential outcomes
- Classic assumption: Unconfoundedness / Missing at Random

$$D_i \perp (Y_i^1, Y_i^0) \quad \forall i \quad (13)$$

- Under the above assumption we can identify

$$ATE = E[\Delta_i] = E[Y_i^1 - Y_i^0]$$

$$\begin{aligned} E[Y_i | D_i = 1] - E[Y_i | D_i = 0] &= E[Y_i^1 | D_i = 1] - E[Y_i^0 | D_i = 0] \\ &= E[Y_i^1] - E[Y_i^0] \\ &= E[Y_i^1 - Y_i^0] \\ &= ATE. \end{aligned} \quad (14)$$

What would OLS identify?

- We can express the potential outcome equation as follows

$$\begin{aligned} Y_i &= Y_i^0 + (Y_i^1 - Y_i^0)D_i \\ &= \mu + \beta D_i + \varepsilon_i \end{aligned} \tag{15}$$

where $\mu = E[Y_i^0]$; $\beta = E[Y_i^1 - Y_i^0]$ and
 $\varepsilon_i = Y_i^0 - E[Y_i^0] + [(Y_i^1 - Y_i^0) - E(Y_i^1 - Y_i^0)]D_i$

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- Does this look familiar?

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- Does this look familiar?
- Just run OLS and you will get back ATE since randomization ensures that $E[D_i \varepsilon_i] = 0$

What would OLS identify?

- We can express the potential outcome equation as follows

$$\begin{aligned} Y_i &= Y_i^0 + (Y_i^1 - Y_i^0)D_i \\ &= \mu + \beta D_i + \varepsilon_i \end{aligned} \tag{15}$$

where $\mu = E[Y_i^0]$; $\beta = E[Y_i^1 - Y_i^0]$ and
 $\varepsilon_i = Y_i^0 - E[Y_i^0] + [(Y_i^1 - Y_i^0) - E(Y_i^1 - Y_i^0)]D_i$

- Does this look familiar?
- Just run OLS and you will get back ATE since randomization ensures that $E[D_i \varepsilon_i] = 0$
- Where is the randomness or uncertainty coming from here?

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- Less assumptions better but can't get something for nothing
- Important to consider implicit assumptions present in both approaches..
- Successful JMPs combine both approaches (complements, not substitutes!).

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- Note that these assumptions are typically embedded in the $f(\cdot, \cdot)$ of a structural model:

$$Y_i = f(D_i, \varepsilon_i)$$

Which if's have causal answers?

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 - “No Causation Without Manipulation!”
- Q2: What is the effect of being perceived to be black (as opposed to white) on wages?
 - Perceptions can be manipulated (e.g. Bertrand and Mullianathan, 2004).
 - Explosion of “Audit Studies” → Another thing “invented” by Economists?

Systematic Discrimination among US Employers

Systemic Discrimination Among Large U.S. Employers

Patrick M. Kline, Evan K. Rose, and Christopher R. Walters

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ABSTRACT

We study the results of a massive nationwide correspondence experiment sending more than 83,000 fictitious applications with randomized characteristics to geographically dispersed jobs posted by 108 of the largest U.S. employers. Distinctively Black names reduce the probability of employer contact by 2.1 percentage points relative to distinctively white names. The magnitude of this racial gap in contact rates differs substantially across firms, exhibiting a between-company standard deviation of 1.9 percentage points. Despite an insignificant average gap in contact rates between male and female applicants, we find a between-company standard deviation in gender contact gaps of 2.7 percentage points, revealing that some firms favor male applicants while others favor women. Company-specific racial contact gaps are temporally and spatially persistent, and negatively correlated with firm profitability, federal contractor status, and a measure of recruiting centralization. Discrimination exhibits little geographical dispersion, but two digit industry explains roughly half of the cross-firm variation in both racial and gender contact gaps. Contact gaps are highly concentrated in particular companies, with firms in the top quintile of racial discrimination responsible for nearly half of lost contacts to Black applicants in the experiment. Controlling false discovery rates to the 5% level, 23 individual companies are found to discriminate against Black applicants. Our findings establish that systemic illegal discrimination is concentrated among a select set of large employers, many of which can be identified with high confidence using large scale inference methods.

Causal effects of choices

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- Often “we” can only encourage these choices.
- Holland: can only talk about effect of encouragement.
- Does this invalidate most of econometrics?
- tl;dr: No! We can impose structure to learn more (+ sometimes it is perfectly ok to only discuss effects of encouragement).
- More on this when we'll talk about IV.